Sharp growth of corner cuts in fixed dimension

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Abstract

Let $C_d(n)$ count the n-point subsets of the nonnegative integer lattice that can be strictly separated from their complements by an affine hyperplane. For every fixed $d \geq 2$, we prove

$$C_d(n) = \Theta_d\big(n^{d-1}(\log n)^{d-1}\big).$$

The upper bound is due to Wagner. For the lower bound, a finite sieve produces primitive lattice simplices in separated axis boxes. Their primitive normal directions are distinguished at a prescribed cardinality by two integer patches on opposite sides of a boundary level. Summing over disjoint aspect ranges supplies the logarithmic factor. This gives a matching lower bound in every fixed dimension, including the three-dimensional order $n^2(\log n)^2$. The cumulative count is consequently of order $n^d(\log n)^{d-1}$. The mean number of completions of a coordinate slice is of order $\log n$, as suggested in Wagner's discussion of the lower bound.

1 The counting problem and its scope

A nonempty finite set $D \subset \mathbb{Z}^d_{\geq 0}$ is a *corner cut* if

$$D = \{x \in \mathbb{Z}^d_{\geq 0} : w \cdot x < t\} \quad \text{for some } w \in \mathbb{R}^d_{>0},\ t > 0,$$

with no lattice point on the cutting hyperplane. Write $C_d(n)$ for the number of such sets of cardinality n, and put

$$A_d(n) = \sum_{j=1}^{n} C_d(j).$$

We count sets, rather than the orders of their elements or the weights representing them. Every cut is an order ideal for the coordinate order. In particular, every coordinate of an element of an n-point cut is at most $n-1$, so these counts are finite.

These objects occur as the exponent sets of initial segments of smooth numbers. For distinct primes $p_1, \ldots, p_d$, comparing the products $\prod_i p_i^{x_i}$ is equivalent to comparing $\sum_i x_i \log p_i$. An exponent prefix is consequently a corner cut. Conversely, every corner cut has such a realization, even with all primes above any prescribed bound (Proposition 7). Thus the full geometric class is exactly the class of exponent-prefix sets as the ordered prime bases vary. This does not give a prime-distribution estimate or a faster random-access algorithm for a fixed prime set.

Corner cuts were introduced in connection with state polyhedra and Gröbner bases by Onn and Sturmfels [1]. The planar order $C_2(n) = \Theta(n \log n)$ is due to Corteel, Rémond, Schaeffer and Thomas [2]. For fixed d, Wagner [3] proved

$$\Omega_d(n^{d-1} \log n) \leq C_d(n) \leq O_d(n^{d-1}(\log n)^{d-1}). \tag{1}$$

His thesis [4, Chapter 3] gives the lower and upper arguments and suggests seeking greater completion multiplicity. Wagner also credits Rémond with independently obtaining the bounds in (1). We prove a matching lower bound, improving the lower logarithmic exponent by $d-2$ when $d \geq 3$. We count primitive normals in each of many aspect ranges and prove that they give distinct prefixes at every sufficiently large prescribed cardinality. The latter step, rather than a lift from a lower dimension, is the key to retaining every logarithmic factor.

Theorem 1. *For each fixed integer $d \geq 2$,*

$$C_d(n) = \Theta_d\big(n^{d-1}(\log n)^{d-1}\big).$$

Here is one explicit lower bound. Put

$$K_d = 4^{d+1}, \qquad T_d = [64d^2(K_d+1)]^d, \qquad R_0(d) = 4d^2(d+1),$$

$$n_*(d) = T_d 4^{R_0(d)}, \qquad \rho_d = 8^{-(d-1)}(100d)^{-d^2},$$

and

$$\lambda_d = \frac{\rho_d}{2^d(4T_d)^{d-1}[4(d+1)]^{d-1}(d-1)!}.$$

For every integer $n \geq n_*(d)$,

$$C_d(n) \geq \lambda_d n^{d-1}(\log_4 n)^{d-1}. \tag{2}$$

Summing gives $A_d(n) = \Theta_d(n^d(\log n)^{d-1})$. The pointwise conclusion matches (1) and strengthens its lower logarithmic exponent when $d \geq 3$. For $d = 2$, it recovers the known planar order.

2 Primitive simplices in axis boxes

Fix an ambient integer $d \geq 2$ and set $\delta = 1/(100d)$. For integer side lengths $L_1, \dots, L_d$, put

$$m = \min_i L_i, \quad M = \max_i L_i, \quad Q = \prod_{i=1}^{d} L_i.$$

Define $B_i \subset \mathbb{Z}^d_{\geq 0}$ by taking the jth coordinate interval to be

$$[L_j, L_j + \lfloor \delta L_j \rfloor] \quad (j = i), \qquad [0, \lfloor \delta L_j \rfloor] \quad (j \neq i).$$

A d-by-$(d-1)$ integer matrix is called primitive if the greatest common divisor of its maximal minors is one.

Lemma 2. *Assume $M \leq m^2$ and $m \geq 2^{8d^2}$. At least a fraction $8^{-(d-1)}$ of the ordered choices $p_i \in B_i$ have primitive difference matrix*

$$V = [p_2 - p_1 \ \cdots \ p_d - p_1].$$

The minor criterion and its equivalence with full column rank modulo every prime are standard [8, Propositions 3–4]. The point requiring proof here is uniformity for these moving, possibly anisotropic boxes.

Proof. Keep the same δ throughout a dimension induction. Let E_r be the event that the first r points, projected onto the first r coordinates, have primitive difference matrix. The event E_1 is automatic, with empty minor equal to one. At step $2 \le r \le d$, fix a primitive configuration of the previous $r-1$ projected points. Its difference matrix U has size $(r-1)$-by-$(r-2)$. Its vector a_0 of signed maximal minors is primitive, and U has rank $r-2$ modulo every prime.

Let u be the first $r-1$ coordinates of $p_r - p_1$, and put $D = \det[U\ u]$. This determinant is nonzero for every allowed choice. Indeed, the normalized row matrix of the previous $r-1$ projected points differs from the identity by at most $(r-1)\delta$ in maximum row-sum norm. The normal of their affine span, normalized to height one, has scaled coordinates within $(r-1)\delta/(1-(r-1)\delta)$ of one and is positive. The new projected point has every scaled coordinate at most δ; it is strictly on the origin side of that affine span. Also,

$$|D| \le (r-1)!\, 2^{r-1} M^{r-1} \le 2^{d^2} m^{2(d-1)}. \tag{3}$$

Every coordinate interval contains more than δm integers. For a prime ℓ, the equation $D \equiv 0 \pmod{\ell}$ is a nontrivial linear equation in the new projected point: at least one coordinate of a_0 is nonzero modulo ℓ. Its probability is at most $1/\ell + a/m$, where $a = 100d$.

Suppose $\ell \mid D$. Since U has full column rank modulo ℓ, there is a unique vector λ with $u \equiv U\lambda \pmod{\ell}$. Write z for the rth row of the old columns and u_r for the rth coordinate of the new column. The full r-by-$(r-1)$ matrix loses rank modulo ℓ precisely if

$$u_r \equiv z\lambda \pmod{\ell}. \tag{4}$$

Conditional on everything except the last coordinate of p_r, this is one residue class, of probability at most $1/\ell + a/m$. Thus the probability of failure at any particular prime is at most $(1/\ell + a/m)^2$.

Put $s = \lfloor (\log_2 m)/2 \rfloor$, $P = 2^s$. The small-prime union bound gives

$$\begin{aligned} \Pr(\text{failure at a prime } \ell \le P) &\le \sum_{\substack{\ell \le P \\ \ell \text{ prime}}} (1/\ell + a/m)^2 \\ &\le \frac{3}{4} + \frac{2a(s+1)}{2^{2s}} + \frac{a^2}{2^{3s}}. \end{aligned} \tag{5}$$

Here $\sum_{j=2}^{\infty} j^{-2} < 3/4$ follows by comparison with $1/4 + \int_2^{\infty} x^{-2}\,dx$. The cross term uses $H_P \le 1 + \log P \le 1 + s$, and the constant term uses at most P primes.

For large primes, first fix the new projected coordinates. Every bad prime $\ell > P$ divides the same nonzero integer D. By (3), their number is at most

$$\frac{\log_2 |D|}{s} \le 4(d-1) + \frac{d^2 + 4(d-1)}{s} \le b := 4d - 3 \quad \text{if } s \ge d^2 + 4(d-1).$$

For each of these finitely many primes, (4) bounds the conditional last-coordinate failure probability by $2^{-s} + a2^{-2s}$. Their total contribution is therefore at most $b(2^{-s} + a2^{-2s})$.

Define

$$E_d(s) = \frac{2a(s+1)}{2^{2s}} + \frac{a^2}{2^{3s}} + b(2^{-s} + a2^{-2s}). \tag{6}$$

This decreases for $s \ge 1$. For every integer $d \ge 2$, the choice $S_d = 4d^2$ satisfies $S_d \ge d^2 + 4(d-1)$ and $E_d(S_d) < 2^{-10} < 1/8$. For an explicit verification valid in every dimension, use $a \le 2^{4d}$, $b \le 2^{2d}$ and $4d^2 + 1 \le 2^{3d}$. They give

$$E_d(4d^2) \le 2^{1+7d-8d^2} + 2^{8d-12d^2} + 2^{2d-4d^2} + 2^{6d-8d^2} < 2^{-10}.$$

The four exponents decrease for $d \geq 2$, and at $d = 2$ are $-17, -32, -12, -20$.

Consequently the conditional probability of a nonprimitive extension is at most $3/4 + E_d(s) \leq 7/8$. To justify the conditioning across steps, the events $E_1, \ldots, E_{r-1}$ depend only on previous projected coordinates. After these are fixed, the old rth coordinates and the new point retain their product distributions. Thus

$$\Pr(E_r \mid E_1, \ldots, E_{r-1}) \geq 1/8.$$

Multiplication proves that $\Pr(E_1 \cap \cdots \cap E_d) \geq 8^{-(d-1)}$. This intersection implies E_d, as required. All bounds on large primes were finite before any limit was taken. □

3 Primitive normals at a prescribed size

For each primitive frame from Lemma 2, orient its cofactor normal a positively and put $H_a = a \cdot p_1$. Let P have rows p_i. Its column-scaled matrix differs from the identity by a matrix of maximum row-sum norm at most $d\delta = 1/100$. The determinant is positive and every eigenvalue lies within $1/100$ of one. Consequently

$$H_a = \det P, \qquad 2^{-d}Q \leq H_a \leq 2^d Q. \tag{7}$$

The solution of $p_i \cdot w = 1$ satisfies $|L_i w_i - 1| \leq d\delta/(1 - d\delta) < 1/10$. Since $w = a/H_a$, this gives

$$\frac{9}{10} < \frac{a_i L_i}{H_a} < \frac{11}{10}. \tag{8}$$

Lemma 3. *Under the hypotheses of Lemma 2, these frames give at least $2^{-d}\rho_d Q^{d-1}$ distinct primitive positive normals.*

Proof. Primitivity gives an integer z with $a \cdot z = 1$. The matrix $[V\ z]$ is unimodular, so the columns of V form a basis of the entire integer kernel of a. If two points of the same B_i lie in the selected affine plane, their difference is Vc, with $c \in \mathbb{Z}^{d-1}$. The submatrix in rows $2, \ldots, d$, after row scaling by the L_i, differs from the identity by at most $(d-1)\delta$ in maximum row-sum norm. Thus

$$\|c\|_\infty \leq \frac{\delta}{1 - (d-1)\delta} < 1,$$

so the two points agree. Each plane determines its selected simplex. Each box has at least $\delta^d Q$ points, so the number of distinct planes is at least $\rho_d Q^d$. For any fixed primitive normal, the integer height in (7) has at most $2^d Q$ possibilities. Division proves the claim. □

When one normal occurs at several heights, fix one witnessing simplex for it, and use that simplex to define H_a. The prefix below depends only on a, not on this choice of witness.

For a positive integer normal a, order lattice points by $(a \cdot x, x_1, \ldots, x_d)$, and let $D_n(a)$ be the first n. Let h be the least integer for which

$$F_a(h) := |\{x \in \mathbb{Z}^d_{\geq 0} : a \cdot x \leq h\}| \geq n.$$

The set $D_n(a)$ is a corner cut, including when the score-h level is only partly included. Indeed $h \leq (n-1)\min_i a_i$, by considering a minimum-weight axis. Every point at score at most h lies in $\{0, \ldots, n-1\}^d$. For $n > 1$, the integer vector $u_i = n^{d-i}$ represents lexicographic order in that box, and $u \cdot x \leq n^d - 1$ there. Taking $w = n^d a + u$ preserves all these levels and strictly orders the boundary level. Positivity keeps every higher a-level above them. A threshold between consecutive integer scores gives a strict representation. The case $n = 1$ is immediate. We will use the weak separation property

$$b \cdot x \leq b \cdot y \quad (x \in D_n(b),\ y \notin D_n(b)). \tag{9}$$

Lemma 4 (Fixed-size injectivity)**.** *Let a, b be primitive positive normals furnished by the same aspect box. Put $K = 4^{d+1}$. If the boundary height h of $D_n(a)$ satisfies*

$$h/H_a \geq 16d^2(K+1), \tag{10}$$

then $D_n(a) = D_n(b)$ implies $a = b$.

Proof. Choose the selected frame $V = [v_1 \ \cdots \ v_{d-1}]$ for a. Since V is a basis of its integer kernel, an integer solution of $a \cdot z = 1$ can be reduced modulo V to give

$$z = p_1/H_a + Vc, \qquad |c_j| \leq 1/2. \tag{11}$$

Each entry satisfies $|(v_j)_i| < 2L_i$, so $|z_i| \leq (d+1)L_i \leq 2dL_i$. Let $y_i = (h-1)/(da_i)$. Reduce $(h-1)z$ modulo V about y, obtaining an integer x_0 such that

$$a \cdot x_0 = h - 1, \qquad x_0 - y = V\beta, \qquad |\beta_j| \leq 1/2.$$

Both integer patches

$$\{x_0 + Vk : k \in \mathbb{Z}^{d-1},\ |k_j| \leq K\}, \qquad \{x_0 + 2z + Vk : k \in \mathbb{Z}^{d-1},\ |k_j| \leq K\} \tag{12}$$

are nonnegative. To see this coordinatewise, the displacement from y_i is less than $(2Kd + 5d)L_i$, while (8) and (10) give

$$y_i/L_i > \frac{h-1}{2dH_a} > 2Kd + 5d.$$

Their a-scores are respectively $h-1$ and $h+1$. Thus the first patch is entirely inside $D_n(a)$, the second entirely outside; the score-h tie is irrelevant.

Assume the two prefixes agree, and put $S = \sum_j |b \cdot v_j|$. By (9), the maximum b-score on the first patch cannot exceed the minimum on the second. Hence

$$KS \leq b \cdot z \leq b \cdot p_1/H_a + S/2. \tag{13}$$

For two normals in the same aspect family, (7) gives $H_b/H_a \leq 4^d$, and (8) gives

$$\frac{b \cdot p_1}{H_a} < 4^d \frac{11}{10}(1 + d\delta) < 2 \cdot 4^d.$$

It follows that $(K - 1/2)S < 2 \cdot 4^d < K - 1/2$. But S is an integer, so $S = 0$. Thus b annihilates every column of V and is parallel to a. Primitivity and positivity force $a = b$. □

Lemma 5. *If $n/Q \geq T_d$ and $m \geq 2^{8d^2}$, then (10) holds. With $G = (d!n/Q)^{1/d}$, the maximum ith axis coordinate of $D_n(a)$ lies strictly between $GL_i/2$ and $2GL_i$.*

Proof. The disjoint half-open unit cubes based at the points counted by $F_a(t)$ contain the positive simplex of height t, and are contained in the simplex of height $t + \sum_i a_i$. Therefore

$$\frac{t^d}{d! \prod_i a_i} \leq F_a(t) \leq \frac{(t + \sum_i a_i)^d}{d! \prod_i a_i} \qquad (t \geq 0).$$

Use $F_a(h-1) < n \leq F_a(h)$ and (8) to get

$$\frac{9}{10}G - \frac{11d}{10m} < \frac{h}{H_a} < \frac{11}{10}G + \frac{1}{H_a}.$$

Since $m \geq 10d$ and $G \geq 64d^2(K_d + 1)$, this implies $4G/5 < h/H_a < 6G/5$, and also (10). If M_i is the axis maximum, all integer axis points below score h are included and none above it are included. Thus $h/a_i - 2 < M_i \leq h/a_i$. Combining these inequalities with (8) gives $GL_i/2 < M_i < 2GL_i$. □

4 Summing aspect ranges

Proof of Theorem 1. The upper bound is Wagner's theorem [3]; see also [4, Theorem 3.15]. For the lower bound, take R maximal with $Q = 4^R \le n/T_d$. For every tuple

$$r_1 + \cdots + r_d = R, \qquad r_i \ge \lceil R/(d+1) \rceil,$$

put $L_i = 4^{r_i}$. When $n \ge n_*(d)$, these tuples satisfy $R \ge R_0(d)$, $\min_i L_i \ge 2^{8d^2}$, and $\max_i L_i \le (\min_i L_i)^2$. Lemmas 3–5 therefore give at least $2^{-d}\rho_d Q^{d-1}$ different n-point cuts per tuple.

The value G in Lemma 5 is common to all tuples. Two distinct tuples differ by a factor of at least four in one L_i. The open observable ranges $(GL_i/2, 2GL_i)$ are disjoint for that coordinate, so no cut is counted twice across tuples.

Put $r = R - d\lceil R/(d+1) \rceil$. The number of tuples is

$$\binom{r+d-1}{d-1} \ge \frac{R^{d-1}}{[2(d+1)]^{d-1}(d-1)!},$$

since $r \ge R/(d+1) - d \ge R/[2(d+1)]$. Also $Q > n/(4T_d)$ and $R \ge (\log_4 n)/2$. For the latter explicit bound, $d^2 \le 4^{d-1}$ gives $T_d \le 4^{2d^2+4d}$, and $R_0(d) \ge 2d^2 + 4d + 2$. Multiplying the counts yields (2). □

5 Consequences and limitations

The theorem also determines the average completion multiplicity raised in Wagner's discussion [4, Section 3.5]. Write $\mathcal{C}_{d-1}(\le n)$ for all nonempty corner cuts of size at most n in dimension $d-1$. For each such cut T, let

$$F_n(T) = |\{D : |D| = n,\ D \text{ a corner cut in } \mathbb{Z}^d_{\ge 0},\ D \cap \{x_d = 0\} = T \times \{0\}\}|.$$

Corollary 6 (Average completion multiplicity)**.** *For every fixed $d \ge 2$,*

$$\frac{1}{|\mathcal{C}_{d-1}(\le n)|} \sum_{T \in \mathcal{C}_{d-1}(\le n)} F_n(T) = \Theta_d(\log n).$$

Proof. Restriction to $x_d = 0$ sends every nonempty corner cut to a nonempty corner cut. Its fibres partition the n-point cuts, so the displayed mean equals $C_d(n)/A_{d-1}(n)$. For $d \ge 3$, Theorem 1 and its cumulative consequence give the numerator and denominator orders $n^{d-1}(\log n)^{d-1}$ and $n^{d-1}(\log n)^{d-2}$, respectively. For $d = 2$, there is one one-dimensional cut of each positive size, so $A_1(n) = n$, giving the same conclusion. □

The restriction, fibre identity and lifting argument are established background [4, Lemma 3.5 and Section 3.5]; the corollary records the consequence of the new matching count. It is a mean over all slices, not a lower bound for each slice. For example, $|T| = n$ forces $F_n(T) = 1$.

There is an established geometric interpretation. For a cut D, put $\alpha_D = \sum_{x \in D} x$. For fixed size n, these vectors are distinct and each is a vertex of their convex hull. Indeed, weights strictly cutting off D make $w \cdot \alpha_D$ smaller than $w \cdot \alpha_E$ for every other n-point set E: each replacement point has greater weight than every removed point. Thus α_D is uniquely exposed. Conversely every vertex of the finite hull is one of its generators. The vertex count of the corner-cut polytope is consequently $C_d(n)$. This correspondence is known [1, 5, 9]; in particular, Müller [5, Lemma 2.6] gives the exposed-vertex argument. Our contribution is the quantitative lower count, not the dictionary. For generic experimental designs the same count describes their corner-cut models [9, Definition 7 and Theorem 2].

Proposition 7 (Realization by prime bases)**.** *For every nonempty finite corner cut $D \subset \mathbb{Z}^d_{\geq 0}$ and every $B > 0$, there are distinct primes $p_1, \ldots, p_d > B$ such that D is the exponent set of the first $|D|$ smooth numbers on these ordered bases.*

Proof. Write $D = \{x : w \cdot x < 1\}$ strictly, and set

$$g = 1 - \max_{x \in D} w \cdot x > 0, \qquad H = \max_{x \in D} \sum_i x_i.$$

Relabel the coordinates temporarily so that $w_1 \leq \cdots \leq w_d$. Choose $T > 0$ with $e^{Tw_1} > \max(B, 2)$ and $dH \log 2/T < g$. Bertrand's postulate [6] supplies a prime in each interval

$$2^{i-1} e^{Tw_i} < p_i < 2^i e^{Tw_i}.$$

These intervals are disjoint, so the primes are distinct, including when some weights agree. Put $v_i = \log p_i / T$. Then $0 < v_i - w_i < d \log 2/T$. For $x \in D$, $v \cdot x < w \cdot x + g \leq 1$, whereas for every $x \notin D$, $v \cdot x \geq w \cdot x > 1$. Consequently the products for $x \in D$ are precisely those below e^T. Unique factorization makes their order strict. Undoing the coordinate relabeling proves the statement. □

For fixed B, let $S_{d,B}(n)$ count distinct exponent sets of length n obtained over ordered tuples of distinct primes greater than B. Count each set once, however many tuples realize it. The proposition and the converse observation in Section 1 give

$$S_{d,B}(n) = C_d(n) = \Theta_d(n^{d-1}(\log n)^{d-1}),$$

$$\sum_{j=1}^{n} S_{d,B}(j) = \Theta_d\big(n^d (\log n)^{d-1}\big).$$

The realization argument is an elementary use of a classical theorem; it is not a separate novelty claim. Its role is to specify exactly which smooth-number shapes the counting result describes. A single fixed prime tuple gives one prefix of each length.

The pointwise count gives a matching asymptotic order in every fixed dimension and for the established polytope and model dictionaries. It supplies no asymptotic leading constant, no new enumeration algorithm, no uniform claim as d grows with n, and no fixed-prime unranking complexity improvement. The prescribed-size step is Lemma 4; the earlier cumulative construction alone did not give that conclusion.

6 Related work and reproducibility

The lower bound concerns the same cardinality count as Wagner's (1). The proof uses three ingredients: a uniform finite primitive-frame estimate, injectivity at a prescribed size, and disjoint aspect ranges. We describe the relevant background for these steps. Müller [5, Sections 2–5] studies the same polytopes, including their planar vertex order and adjacency. Her enumeration discussion [5, Section 1, p. 324] cites the planar result and Wagner's bounds; her structural theorems do not supply the bounded-cardinality count proved here.

The unrestricted density of rectangular primitive integer matrices is known [7, 8]. The later paper repairs a defect in the earlier density proof. A centered-cube density alone does not assert the uniform estimate for our moving boxes; Lemma 2 gives a direct finite proof. We claim no new minor criterion or unrestricted density formula.

Chen, Feng, Liu and Wu [13, Theorem 1] give finite probability bounds for appending independent rows to a prescribed primitive matrix. Their random entries use a common

interval of length at least the maximum prescribed entry. Our boxes have translated intervals of different lengths, with their shortest length potentially much smaller than the largest entry. That theorem is not a direct substitute for Lemma 2. Prime-by-prime rank analysis and finite determinant cutoffs are established methods; our claimed contribution is their use in the multiscale corner-cut count. The additional fixed-size step is the two-patch comparison in Lemma 4.

Here is a precise link with finite-grid threshold functions. Put $G_n = \{0, \dots, n\}^d$. An n-point corner cut is exactly an n-set of G_n containing the origin, where an n-set is a subset strictly separable from its complement in G_n. The forward implication follows from the coordinate bound in Section 1. For the converse, write a separator as $w \cdot x < t$, with $t > 0$ since the origin is included. If $w_i \le 0$, all $n+1$ points $0, e_i, \dots, ne_i$ are included, a contradiction. Thus $w_i > 0$. Also ne_i is excluded for every i, so $nw_i > t$; every nonnegative lattice point outside G_n has score greater than t. This proves the converse. Wagner [4, Section 3.2] uses a smaller finite ground set for enumeration; the finite-set reduction itself is not new.

The counting parameter matters even after this identification. Haukkanen and Merikoski [11, Equations (1) and (9)] count all threshold dichotomies of an m-by-r grid, obtaining

$$t(m, r) = \frac{6}{\pi^2}(mr)^2 + O(mr^2) \qquad (2 \le m \le r).$$

This total sums over every possible included cardinality. In particular, $t(n+1, n+1) = \Theta(n^4)$, while already $C_2(n) = \Theta(n \log n)$. A lower estimate for the total does not give a lower estimate for the selected n-point class at the origin. Lemma 4 supplies the prescribed-size control in our construction; the known total-grid count is not being presented as a new result.

Constraining a normal by points on parallel boundary levels also has precedents in digital-plane recognition. Coeurjolly et al. [12, Proposition 6 and Theorems 8 and 11] describe feasible plane parameters using upper and lower leaning polygons, under projection or connected-slice hypotheses. Our argument requires the quantitative conclusion that bounded primitive integer normals are distinguished at every sufficiently large prescribed cardinality. We do not claim the general principle of boundary-point recognition as new.

The Hilbert zonotope refines state polyhedra and supports polynomial-time computation of universal Gröbner bases in fixed dimension [10, Proposition 3.2 and Theorem 3.5]. Its count is an upper bound for a common refinement, not the lower bounded-cardinality count proved here.

Elizalde and Galván discuss higher-dimensional pyramidal partitions and refer to Wagner's growth bounds [15, Section 8.2]. Our comparison uses the preprint version [15]; the journal article is listed separately as [16].

Kalka and Kuhnt [14, Section 2.2, p. 6] explicitly restate Wagner's lower and upper orders for corner cuts, while treating a numerical statistical fan. That passage does not supply a stronger corner-cut counting bound or a proof of our prescribed-size lemma.

Scope of the comparison. The count in Theorem 1 improves the lower bound of [3, 4] and matches their upper bound. The restriction and lifting viewpoint, primitive-matrix methods, boundary-point recognition, and polytope and model correspondences are attributed above. The improvement is the resulting uniform cardinality count. The accompanying research audit records precisely which source versions and statements were inspected, including the use of the author's thesis proofs for [3] and the deferred journal-version check for [16]. It is available at `results/research/corner_cut_growth_audit.json`.

Exact checks. The standard-library Python checker `scripts/run_corner_cut_growth_checks.py` uses independent determinants, modular Gaussian elimination and exact ratio-

nal feasibility. It checks primitive-plane injectivity, the conditional residue bound and a nonprimitive-projection negative control. The prescribed-size checks construct integer Bezout increments and affine-coset reductions, then verify reversed orders on the two patches for 2,280 ordered pairs of distinct normals in dimensions two through six, each at five boundary heights with different residues relative to the witness height. All 11,400 pair–height checks pass. The exact points and scores are saved. The 200 tie-resolution fixtures include 150 partial boundary levels and 50 complete ones. It exhausts stated small boxes in dimensions two through four and uses deterministic samples in dimensions five and six. Six prime-realization fixtures, including equal weights and unused coordinates, are checked by integer products and an independent heap generation of the corresponding smooth-number prefixes. The explicit constants and 352 scale-rounding cases in dimensions two through twelve are checked with rational arithmetic. These computations can falsify steps; they do not prove the uniform theorem or its novelty.

Run, from the repository root,

```
python3 -B scripts/run_corner_cut_growth_checks.py
```

The report records fixtures, witnesses, the exact command and SHA-256 hashes of its sources in `results/research/corner_cut_growth_checks.json`. The proof above is independent of the computation. The other project kernels and public rank/unrank interface are unchanged.

Assistance and review. OpenAI Codex proposed and developed the quantitative argument, carried out the literature comparison, and produced the computational checks and manuscript draft during an investigation prompted by the project owner. The argument has undergone internal review; no independent referee assessment or journal acceptance is claimed. The proof is included in full so that it can be checked independently of the software.

References

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